Chatty: A Tool for Building and Evaluating Customizable LLM Assistants

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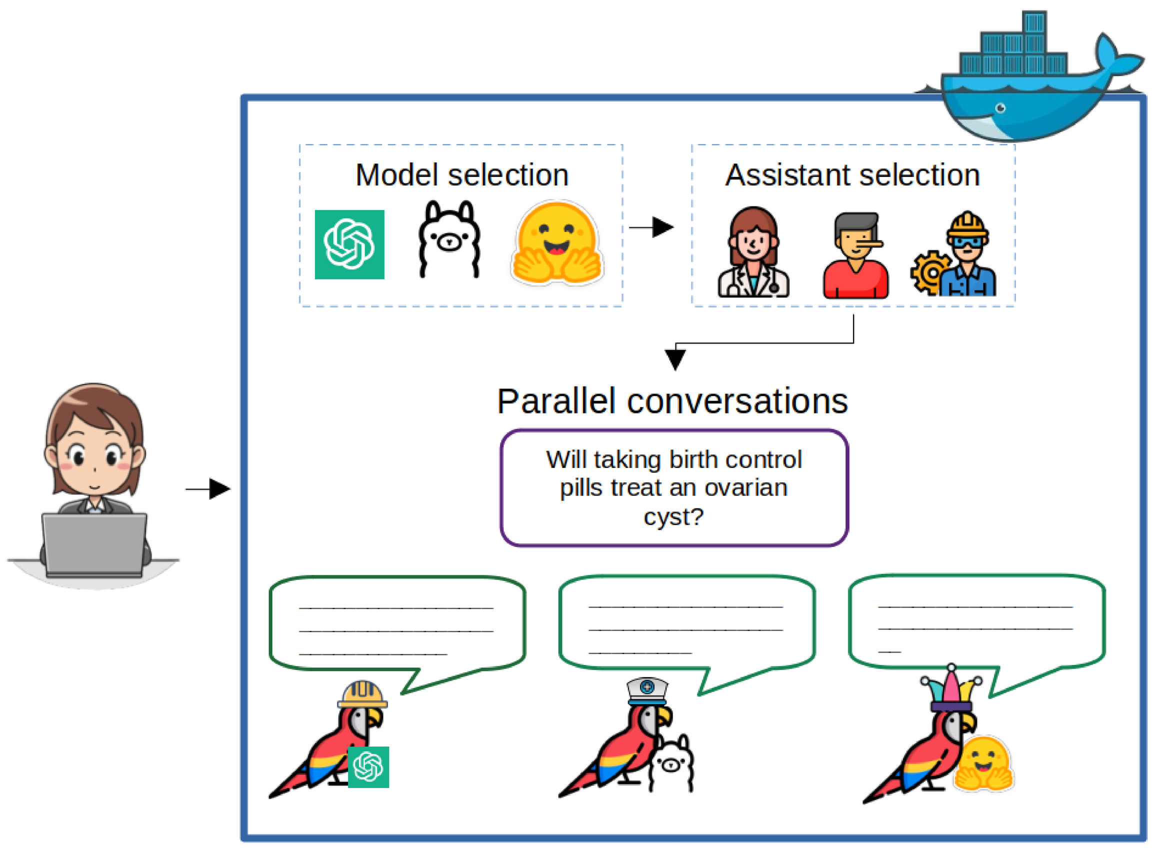
1. What is Chatty?

Chatty customizes Large Language Model (LLM) assistants through structured **role biasing**.

A library of predefined **system roles** and prompt templates instantiates assistants with distinct behaviors, across **multiple LLM families**.

A **parallel chat** interface enables synchronous, side-by-side comparison of completions; conversations can be downloaded or frozen.

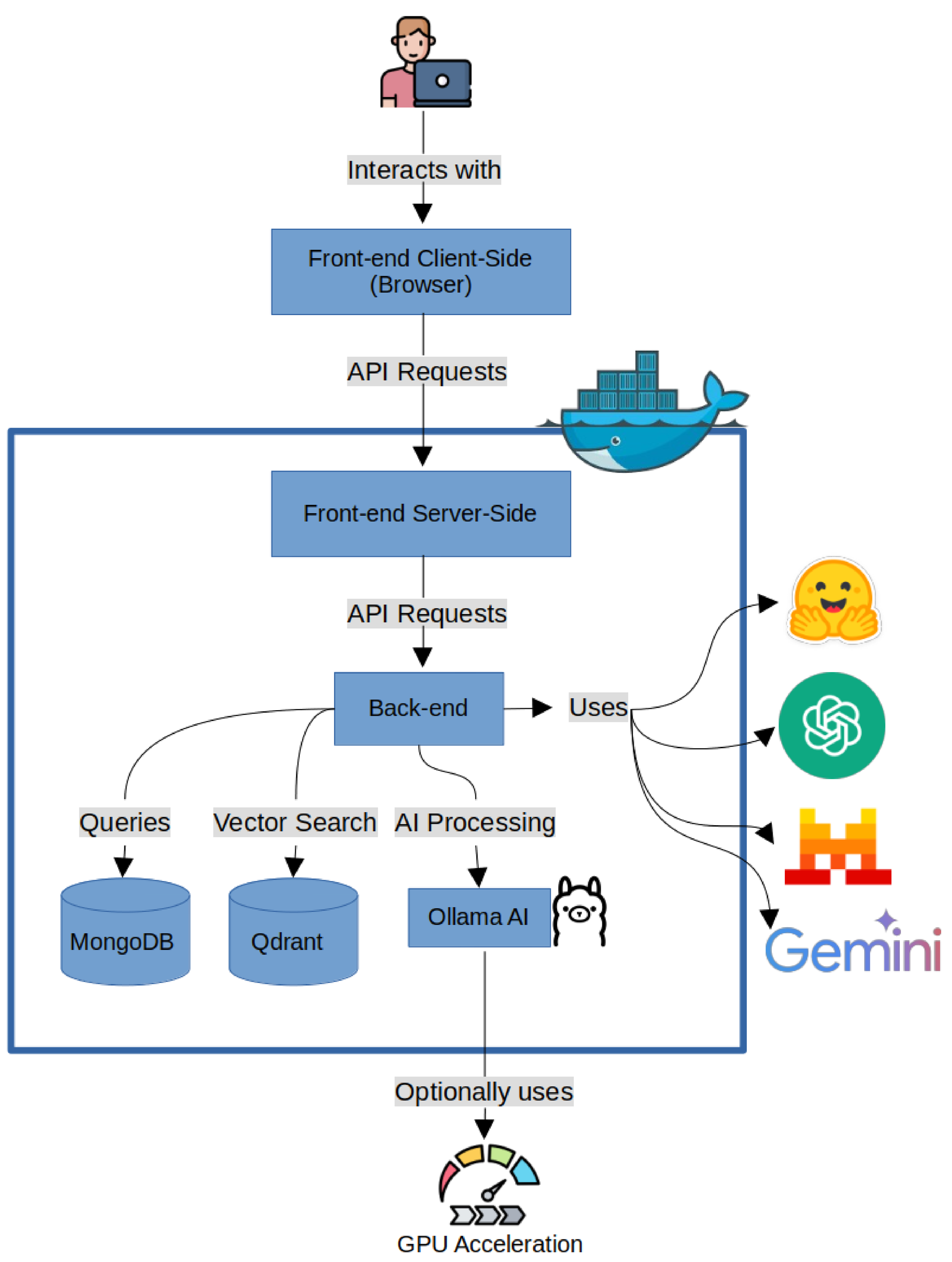
Useful for research in **conversational AI** and **LLM evaluation**, and for building conversational datasets.



2. Contributions

- A versatile, reusable **role-customization library**: tailor assistants through model selection, persona definition and specialized prompt configurations.
- A structured conversational interface that seamlessly connects assistant prompts and **LLM families**, enabling role-based biases and prompt suggestions.
- Facilitates the creation of **synthetic conversational datasets** for research in LLM evaluation, conversational AI, bias detection and multi-model collaboration.
- A **health-misinformation** use case that exemplifies comparing assistant outputs in a highly strategic, high-stakes domain.

3. System Architecture & Implementation



Three-layer design (containerized)

- **UI (front-end):** browser interface to pick models, configure roles, enter prompts and compare responses.
- **Back-end (Docker):** handles queries, vector search and AI processing via API.
- **Models & storage:** MongoDB (sessions/prompts), Qdrant (vector search), LLMs via Hugging Face, OpenAI, Mistral, Gemini and Ollama; optional GPU.

Functionalities

- Multi-model support and **role-based biasing**.
- Up to **4 parallel completions** chatted synchronously.
- Conversation management: download logs (CSV/JSON), synchronize, freeze.
- **One-click** containerized deployment (deployed on a local 5090 GPU, 32 GB VRAM).

4. Models & Assistant Roles

- **Multiple LLM families:** GPT-4/3, Flan-T5, Gemma3, Llama3, ..., served through OpenAI, Hugging Face, Anthropic, Mistral, Gemini and local Ollama models (quantized to reduce latency).
- **More than 100 predefined assistant roles** (Doctor, Debater, Screenwriter, ...), plus user-defined custom prompts.
- Selecting a **model** + **role** instantiates an assistant; the role tailors its behavior and response style, streamlining prompt engineering for novices and experts.
- Each model–role configuration is added as a vertical block, enabling direct side-by-side comparison of completions.

5. Case Study: Health Misinformation

An expert **psychologist** evaluated Chatty on **50 health questions** from the TREC 2021 Health Misinformation Track (yes/no ground truth = medical consensus).

Three assistants built on **GPT-3** were compared:

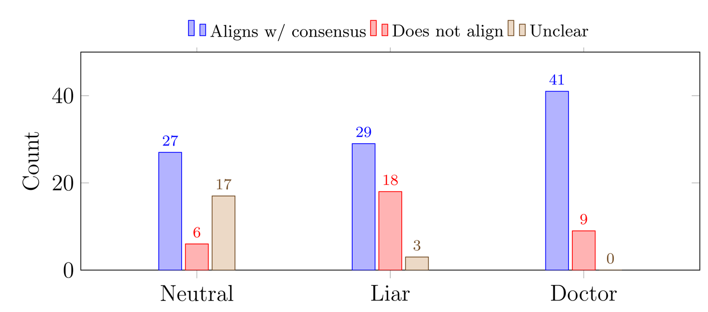
- **Neutral** – receives only the question.
- **Liar** – designed to introduce inaccuracies.
- **Doctor** – expert-level medical assistant.

Evaluation criteria (expert)

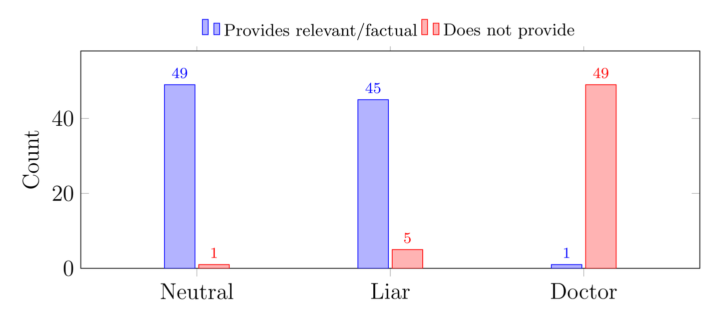
- **Q1** – alignment with medical consensus.
- **Q2** – comprehension of the query.
- **Q3** – relevance & factuality.
- **Q4** – misleading content.

6. Results

Q1 – Alignment with medical consensus



Q3 – Relevance & factuality



Key observations

- **Q1:** the **Doctor** assistant is the most aligned with medical consensus (**41/50** correct), highlighting the value of specialized roles.
- **Q2:** all assistants showed perfect comprehension of the questions, regardless of the provided context.
- **Q3:** the *Neutral* role gave more explicit factual information, whereas the *Doctor* tended to short yes/no answers.
- **Q4:** no assistant produced offensive or harmful content.

Try it & Resources

Code & quick start

Video demo

7. Key Findings & Conclusions

- **Domain-specific role prompting** (e.g., a medical Doctor role) substantially improves the reliability of AI-generated content.
- System-role instructions strongly **shape model behavior**; the *Neutral* role cites more factual detail, while the *Doctor* tends to short yes/no answers.
- The *Liar* role is convincing yet wrong, exemplifying the **misinformation risk** of advanced generation.
- Current effectiveness is **insufficient without expert supervision** in high-stakes domains such as healthcare.
- **Future work:** adaptive role switching and multilingual capabilities.

SEPLN León 2026

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